

Micro Syllabus

Subject: Electromagnetics II/II

Marks Allotment	Theory	Practical	Total
Sessional	40		40
Final	60		60
Total	100	25	125

Time Allotment	Lecture	Tutorial	Practical
Weekly Hours			
Depth Codes	C – Circuit	D – Definition	Dm – Demonstration
Dr – Derivation	Dw – Drawing	E – Explanation	I – Illustration
N – Numerical	P – Proof	Pr – Programming	S – State

Unit	Topic/ Sub topic	Depth Code	Description of Depth	Actual Plan		Plan for this semester		Week
				L	T	L	T	
1	Introduction			3	1			1
	1.1. Scalar and vector fields	D, E,	Scalars and Vectors definition with examples	0.25				
	1.2. Operations on scalar and vector fields.	D, E, I	Vector Algebra, Position vectors, dot product and cross product	0.75				
	1.3. Co-ordinate systems (Cartesian, Cylindrical and Spherical) and conversions	D, E, I, Dr, P, N	Vector Components and Unit Vectors, point transformation and Vector Transformation derivation of tables with few vector transformation examples.	2	1			

2	Electric field			15	5			2,3,4,5,6
	2.1 Coulomb's law 2.2 Electric field intensity	D, E, I, Dr, P, N	The Experimental Law of Coulomb, first law and second law, Electric Force, Electric Field Intensity definition, Field due to different charge distribution namely point, line, sheet and volume charge distribution their derivation and numerical Streamlines and sketches with numerical	3	1			
	2.3 Electric flux density 2.4 Gauss's law and applications 2.5 Physical significance of divergence, Divergence theorem	D, E, I, Dr, P, N	Electric Flux density definition and relation with electric field intensity, Gauss law in integral form and some application in symmetrical charge distribution like point charge, line charge and coaxial cable, Gauss law in unsymmetrical case(differential volume element) and derivation of gauss law in point form, Maxwell first equation and divergence theorem	2	1			
	2.6 Electric potential, potential gradient	D, E, I, Dr, P, N	Energy expended in moving a point charge in an electric field, definition of potential difference and absolute potential, potential field due to point charge, line charge and sheet of charge with numerical, Conservative property, potential gradient and its numerical, definition of electric dipole	2	1			

2.7	Energy density in electrostatic field		energy density in electrostatic field its derivation and numerical	1	0.5			
2.8	Electric properties of material medium	D, E, I, Dr, P	Conductor properties and boundary condition, semiconductors conductivity, nature of dielectric materials with proper explanation of free and bound charges, polarization, electric susceptibility, relative permittivity	3	0.5			
2.9	Free and bound charges, polarization, relative permittivity, electric dipole		boundary condition of perfect dielectric materials with numerical. Capacitance, several capacitance examples (parallel plate capacitor, coaxial transmission line, spherical capacitor, isolated sphere with dielectric coating, two layer dielectric capacitor, two wire line, capacitance of a cylinder above a plane) with their capacitance derivation.					
2.10	Current, current density, conservation of charge, continuity equation, relaxation time	D, E, I, Dr, P, N	Current and current density, convection current density, continuity equation of current integral form and point form with numerical, Metallic conductor and relation between current density and electric field, ohms law in point form and relaxation time constant derivation.	2	0.5			
2.11	Boundary value problems, Laplace and Poisson equations and their solutions, uniqueness theorem	D, E, I, Dr, P, N	Derivation of Laplace and Poisson's Equation, Uniqueness theorem, Use of Laplace equation to find capacitance of parallel plate capacitor, coaxial transmission Line, angled plate geometry, concentric sphere geometry, spherical capacitor, cone suspended above a conducting plane	2	0.5			

3	Magnetic field			8	4			7,8, 9,10
3.1.	Biot-Savart's law	D, E, I, Dr, P, N	Biot-Savart law, Ampere's Circuit law and its application on infinitely long filament carrying current I, coaxial cable, sheet of charge.	2	0.5			
3.2.	Magnetic field intensity							
3.3.	Ampere's circuital law and its application							
3.4.	Magnetic flux density	D, E, I, Dr, P, N	Magnetic Flux and magnetic flux density, Curl, Stoke's Theorem with few numerical	1	0.5			
3.5.	Physical significance of curl, Stoke's theorem							
3.6.	Scalar and magnetic vector potential	D, E, I, Dr, P, N	Scalar and magnetic vector potential derivation and numericals	0.5	0.5			
3.7.	Magnetic properties of material medium	D, E, I, Dr	Different magnetic materials and their properties. Diamagnetic, Paramagnetic, Ferromagnetic, Antiferromagnetic, Ferrimagnetic, Superparamagnetic.	0.5				

3.8. Magnetic force, magnetic torque, magnetic moment, magnetic dipole, magnetization	D, E, I, Dr, P, N	Force on a moving charge, force on a infinitesimal current element, hall effect and hall voltage, force and torque on a closed circuit. Magnetization and Permeability,	1.5	1				
3.9. Magnetic boundary condition	D, E, I, Dr, P, N	Magnetic boundary condition derivation and numerical, the total energy for the steady magnetic field	0.5	0.5				

4	Time Varying Fields (4 hours)			3	1			11
4.1. Faraday's law, transformer emf, motional emf.	D, E, I, Dr, P, N	Faraday law, Lenz's Law, emf due to motional and time varying field, Ampere's law and continuity equation ambiguity, Displacement current, Maxwell's equations in integral form and point form.	3	1				
4.2. Displacement current								
4.3. Maxwell's equations in integral and point forms								

5	Plane Waves 9 hours			9	3			12,13,14
5.1. Wave propagation in lossless and lossy dielectric	D, E, I, Dr, P, N	Maxwell equations in lossy and lossless mediums, plane waves in free space and phasor forms, wave propagation in free space, lossless dielectric and good conductors their derivation and numerical, Intrinsic impedance, velocity, propagation constants, wave impedance, skin effect, skin depth, loss tangent, displacement current density vs conduction current density,	5	1				
5.2. Plane waves in free space, lossless dielectric, good conductor								
5.3. Power and poynting theorem, Average Power Density.	D, E, I, Dr, P, N	Poynting theorem and wave power, time average power density and their numerical in wave propagation	1	1				
5.4. Reflection of plane wave at normal incidence	D, E, I, Dr, P, N	Reflection of plane wave with normal incidence its derivation for reflection coefficient, transmission coefficient, standing waves and maximum amplitude points for electric and magnetic field, standing wave ratio and input intrinsic impedance.	3	1				
5.5. Standing Wave and SWR								
5.6. Input Intrinsic Impedance								
6	Transmission lines (4 hours)			2	2			15
6.1 Transmission line equations (Taking analogy from Wave equations)	D, E, I, Dr, P, N	Types of Transmission line few examples, transmission line parameters, transmission line equations, characteristics impedance and propagation constant. Lossless and Distortion less lines and their propagation constant and characteristic impedance. Reflection coefficient, VSWR and input impedance and their calculation in transmission line.	2	2				
6.2 Lossless, Lossy and Distortionless transmission lines								
6.3 Input impedance, reflection coefficient, standing wave ratio								

Evaluation Scheme:

There will be 11 Questions covering all the chapters in the syllabus. The evaluation scheme for the questions will be indicated in the table below:

Unit	Hour	Number of Questions	Mark distribution
1. Introduction	4	1	5
2. Electric field	15	4	20
3. Magnetic Field	9	2	12
4. Time Varying Fields	4	1	6
5. Plane Waves	9	2	12
6. Transmission Line	4	1	5
Total	45	11	60

Tutorial

1. Numerical related to conversion of coordinate systems (Cartesian to cylindrical /spherical and vice versa , cylindrical to spherical and vice versa)
2. Numerical related to electric field intensity and flux density (Coulomb's law, Gauss Law, Divergence, electric potential and energy density)
3. Numerical related to boundary condition, electric dipole, and boundary value problems.
4. Numerical related to Magnetic Fields (Biot-Savart law, Ampere Circuit law, Stoke's Theorem, magnetic force and torque)
5. Numerical related to time varying fields (transformer/motional emf, displacement current)
6. Numerical related to wave propagation equations in lossy and lossless medium (Poynting theorem, standing wave ratio and intrinsic impedance)
7. Numerical related to Transmission line (lossless, lossy and distortionless)

Laboratory

1. Teledeltos (electro-conductive) paper mapping of electrostatic fields
2. Determination of dielectric constant, display of a magnetic Hysteresis loop
3. Studies of wave propagation on a lumped parameter transmission line
4. Microwave sources, detectors, transmission lines
5. Standing wave patterns on transmission lines, reflections, power patterns on transmission lines, reflections, power measurement
6. Familiarizations of Electric and Magnetic field measurements using simulation tool.

Reference Book:

- W. H. Hayt, "Engineering Electromagnetics", McGraw-Hill Book Company
- J. D. Kraus, "Electromagnetics", McGraw-Hill Book Company
- N. N. Rao, "Elements of Engineering Electromagnetics", Prentice Hall
- Devid K. Cheng, "Field and Wave Electromagnetics", Addison-Wesley
- M. N. O. Sadiku, "Elements of Electromagnetics", Oxford University Press.